

NONNEGATIVE-DEFINITE INDEPENDENCE DISTRIBUTION-PRESERVING COVARIANCE STRUCTURES FOR THE SAMPLE COVARIANCE MATRIX

Dean M. Young

Institute of Statistics, Baylor University, Waco, TX 76798-8005

Laura A. Thompson

Department of Mathematical Sciences, University of Houston–Clear Lake,
Clear Lake, TX 77058

and

Danny W. Turner

Department of Mathematics, Winthrop University, Rock Hill, SC 29733-0001

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Summary: We explicitly characterize the general mean matrix and the general nonnegative-definite covariance structure of a matrix-normal random matrix such that the multivariate sample covariance matrix is distributed as a central Wishart random matrix. We also characterize the general mean matrix and the general nonnegative-definite covariance structure of a matrix-normal random matrix such that the sample covariance matrix is distributed as a central Wishart random matrix and is independent of the sample mean vector.

1. Introduction

Ignoring requisite i.i.d. assumptions for sampled observations in statistical inference problems can yield severe consequences. However, for some statistics one can have a non-i.i.d. covariance structure for the sampled observations without affecting the sampling distributions. That is, there

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sometimes exist observation dependency structures for which the sampling distribution of a statistic is the same as that under the assumption of i.i.d. observations. We refer to such covariance structures as *independence distribution preserving* (IDP).

IDP dependency structures yield insight into a type of model misspecification robustness for a statistic of interest. The existence of IDP covariance structures for a statistic implies that normally distributed observations need not be independent nor must their marginal covariance structures be equal in order for the usual i.i.d.-induced properties of the statistic to hold. Hence, for certain statistics, some degree of robustness against dependent observations with non-identical marginal covariance matrices exists.

In this paper we derive two IDP covariance structure characterizations for the sample covariance matrix. First, we characterize the general n.d. IDP covariance structure of a matrix-normal random matrix such that the sample covariance matrix is distributed as a central Wishart random matrix with $n - 1$ degrees of freedom. Our first result extends the work of Baldessari (1965), Baldessari and Gallo (1981), Stadje (1984) and Jensen (1989), who have characterized the p.d. IDP covariance structure for the univariate sample variance. The general n.d. IDP covariance structure characterization also simplifies a result of Young, Lehman and Meaux (1966), who have characterized the general n.d. independence covariance structure for the univariate sample variance. Moreover, the new IDP result generalizes the sufficient p.d. IDP covariance structure for the sample covariance matrix given by Morrison (1962) and also generalizes the p.d. IDP covariance characterization results of Pavur (1987) and Young, Meaux and Seaman (1994) to the n.d. case.

Second, we characterize the general n.d. IDP covariance structure of a matrix-normal random matrix such that the sample covariance matrix is distributed as a central Wishart random matrix with $n - 1$ degrees of freedom and is independent of the sample mean vector. This result extends the work of Bhat (1962), Rogers (1980) and Stadge (1984), who have given the general p.d. IDP covariance structure such that a multiple of the univariate sample variance is distributed as a central chi-squared random variable with $n - 1$ degrees of freedom and is independent of the univariate sample mean. Our second result also generalizes the work of Khatri (1963), who restricted the form of the covariance structure, and generalizes the results of Pavur (1987) and Young, Meaux and Seaman (1994) from the p.d. to the n.d. IDP covariance structure case.

Before developing our two main IDP covariance-matrix characterization results, we present notation and preliminary lemmas in Section 2. We derive our two characterizations of the matrix-normal IDP general dependency structures in Section 3. We conclude with some brief comments in Section 4.

2. Mathematical preliminaries

In this section we specifically define the two basic problems we address and present definitions and notation that we utilize throughout the remainder of the paper. In addition, we state five lemmas that we use to prove the main results.

To define the two IDP characterization problems for the sample covariance matrix more precisely, let $\mathbf{Y}_{n \times p} \sim \text{MN}(\boldsymbol{\mu}, \mathbf{V})$, where $\boldsymbol{\mu}$ is an $n \times p$ mean matrix and $\mathbf{V}$ is an $np \times np$ n.d. covariance matrix. Also, let $\mathbf{I}_n$ be the $n \times n$ identity matrix, let $\mathbf{M}$ be an $n \times n$ matrix of ones, and let $\mathbf{Y}'$ be the transpose of $\mathbf{Y}$. Define the sample covariance matrix to be

$$\mathbf{S} \equiv \frac{\mathbf{Y}' \left[\left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \right] \mathbf{Y}}{n - 1}, \quad (1.1)$$

and let $W_p(n-1, \Sigma)$ denote a central Wishart distribution with $n-1$ degrees of freedom with the $p \times p$ parameter matrix Σ , where Σ is a nonzero n.d. matrix. We remark that if $Y_{n \times p} \sim MN(\mu, Y)$, where $V = I_n \otimes \Sigma$, then $S \sim W_p(n-1, (n-1)^{-1} \Sigma)$.

We utilize the following notation. The symbol $\mathbb{C}_{m \times n}(\mathbb{R}_{m \times n})$ represents the vector space of all $m \times n$ matrices over the complex (real) field $\mathbb{C}(\mathbb{R})$. We use $\mathbb{C}_n^{\geq}(\mathbb{R}_n^{\geq})$ to represent the cone consisting of all n.d. matrices in $\mathbb{C}_{n \times n}(\mathbb{R}_{n \times n})$, and we define the matrix $\Sigma^{\frac{1}{2}} \in \mathbb{R}_n^{\geq}$ to be the unique square-root matrix of $\Sigma \in \mathbb{R}_n^{\geq}$. The symbol A^* denotes the conjugate transpose of $A \in \mathbb{C}_{m \times n}$. We denote the Moore-Penrose inverse of $A \in \mathbb{C}_{m \times n}(\mathbb{R}_{m \times n})$ by A^+ , and we represent a conditional or $\{1\}$ inverse of $A \in \mathbb{C}_{m \times n}(\mathbb{R}_{m \times n})$ by A^- .

We use the notation $\mathcal{N}(\cdot)$ and $\mathcal{R}(\cdot)$ to represent the null space and range space, respectively. The notation χ_k^2 represents a central chi-squared random variable with k degrees of freedom, and the notation e_n denotes an $n \times 1$ vector of ones.

We now give five preliminary lemmas that we use in the derivation of our two main IDP characterization results. The first lemma is sometimes referred to as the left-hand cancellation rule. One can find a proof in Theorem 2 of Marsaglia and Styan (1974).

Lemma 1. Let $A \in \mathbb{R}_{m \times n}$, $B \in \mathbb{R}_{n \times p}$, $c \in \mathbb{R}_{p \times q}$, $D \in \mathbb{R}_{p \times q}$. If $ABC = ABD$ and $\text{rank}(AB) = \text{rank}(B)$, then $BC = BD$.

The following lemma states a relationship of two particular matrix equations, given certain conditions on the ranks of the matrices.

Lemma 2. Let $A, B \in \mathbb{R}_{n \times n}$ such that $\text{rank}(A) = \text{rank}(BA)$. Then, $BABA = BA$ if and only if $ABA = A$.

Proof. The necessity portion of the lemma follows directly from Lemma 1, and the sufficiency portion is trivial.

The third lemma gives conditions for the existence of a solution and a representation of the general n.d. solution to the matrix equation $AXA^* = B$. One can find a proof in Theorem 1 in Baksalary (1984).

Lemma 3. Let $A \in \mathbb{C}_{m \times n}$ and $B \in \mathbb{C}_m^{\geq}$ such that $\mathcal{R}(B) \subseteq \mathcal{R}(A)$. A representation of the general n.d. solution to the equation $AXA^* = B$ is $X = A^+D + [(I_n - A^+A)Z][A^+D + (I_n - A^+A)Z]^*$, where $D \in \mathbb{C}_{m \times n}$ is an arbitrary but fixed matrix such that $B = DD^*$ and $Z \in \mathbb{C}_{n \times n}$ is arbitrary.

The next lemma gives necessary and sufficient conditions on the parameter matrices of a matrix-normal distribution for a family of multivariate quadratic forms to be distributed as independent noncentral Wishart random matrices. The lemma is a restatement of Theorem 3.4 of Wong and Wang (1993).

Lemma 4. Let $Y_{n \times p} \sim MN(\mu, V)$, where $\mu \in \mathbb{R}_{n \times p}$ and $V \in \mathbb{R}_{np}^{\geq}$, $A_i \in \mathbb{R}_n^{\geq}$, and $\Sigma \in \mathbb{R}_p^{\geq}$ such that $\Sigma \neq 0$. Then, $Y'A_iY \sim W_p(m, \Sigma, \Delta)$ and $Y'A_iY$ is independent of $Y'A_jY$, $i, j = 1, 2, \dots, k$, $i \neq j$, if and only if for some $W \in \mathbb{R}_n^{\geq}$,

$$(1) (A_i \otimes I_p) V (A_i \otimes I_p) = (A_i W A_i) \otimes \Sigma,$$

$$(2) W A_i W A_i = W A_i \text{ and } \text{rank}(W A_i) = m,$$

$$(3) \mu' A_i W A_i \mu = \mu' A_i \mu = \Delta, \text{ and}$$

$$(4) (A_i \otimes I_p) V (A_j \otimes I_p) = 0.$$

Also, $Y'A_iY \sim W_p(m, \Sigma)$ if and only if (1) and (2) hold and $A_i\mu = 0$.

We remark that (4) is a necessary and sufficient condition for this family of matrix-normal based multivariate quadratic forms to be independent.

The last lemma gives conditions for the existence of a n.d. solution and a representation of the general n.d. solution to a particular linear matrix equation. One can find a proof of the lemma in Khatri and Mitra (1976).

Lemma 5. Let $A, C \in \mathbb{C}_{m \times n}$. Then $AX = C$ has a n.d. solution if and only if $CA^* \in \mathbb{C}_m^{\geq}$ and $\text{rank}(CA^*) = \text{rank}(C)$. Provided a solution exists, the general n.d. solution is of the form $X = C^*(CA^*)^- C + (I_n - A^- A)U(I_n - A^- A)^*$, where $(CA^*)^-$ and A^- are arbitrary conditional inverses of CA^* and A , respectively, and $U \in \mathbb{C}_n^{\geq}$ is arbitrary.

3. The main results

We now present a preliminary theorem from which we obtain the first of our two IDP covariance structure characterization results for the sample covariance matrix. Assuming $Y_{n \times p} \sim MN(\mu, V)$, we derive a characterization of the general n.d. IDP covariance structure $V \in \mathbb{R}_{np}^{\geq}$ such that $S \sim W_p(n-1, (n-1)^{-1}\Sigma, \Delta)$. That is, we characterize the general n.d. IDP covariance structure of a matrix-normal random matrix such that S , defined in (1.1), is distributed exactly as if the observations have the i.i.d. covariance structure $V = I \otimes \Sigma$. We first prove the following theorem.

Theorem 1. Let $Y_{n \times p} \sim MN(\mu, V)$, where $\mu \in \mathbb{R}_{n \times p}$ and $V \in \mathbb{R}_{np}^{\geq}$, and let S be defined as in (1.1). Then, $S \sim W_p(n-1, (n-1)^{-1}\Sigma, \Delta)$, where $\Sigma \in \mathbb{R}_p^{\geq}$ and $\Delta = (n-1)^{-1}\mu'(I_n - \frac{1}{n}M)\mu$, if and only if

$$V = \left(I_n - \frac{1}{n}M\right) \otimes \Sigma + (M \otimes I_p)H + H'(M \otimes I_p), \quad (3.1)$$

where $\mathbf{H} = \frac{1}{n} \mathbf{Z} \left[(\mathbf{I}_n - \frac{1}{n} \mathbf{M}) \otimes \boldsymbol{\Sigma}^{\frac{1}{2}} + \frac{1}{2} \mathbf{Z}' \left(\frac{1}{n} \mathbf{M} \otimes \mathbf{I}_p \right) \right]$ with $\boldsymbol{\Sigma} \in \mathbb{R}_p^{\geq}$ and $\mathbf{Z} \in \mathbb{R}_{np \times np}$ is arbitrary.

Proof. To prove sufficiency, assume that $\mathbf{V}$ is defined as in (3.1). Consider the pair of matrix equations

$$\mathbf{W} \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \mathbf{W} \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) = (n-1) \mathbf{W} \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \quad (3.2)$$

and

$$\boldsymbol{\mu}' \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \mathbf{W} \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \boldsymbol{\mu} = (n-1) \boldsymbol{\mu}' \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \boldsymbol{\mu}, \quad (3.3)$$

where $\mathbf{W} \in \mathbb{R}_p^{\geq}$ and $\text{rank} [\mathbf{W} (\mathbf{I}_n - \frac{1}{n} \mathbf{M})] = (n-1)$. Note that from Lemma 2, equations (3.2) and (3.3) hold simultaneously if

$$\left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \mathbf{W} \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) = (n-1) \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right). \quad (3.4)$$

A common n.d. solution to (3.2) and (3.3) is

$$\mathbf{W} = (n-1) \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right). \quad (3.5)$$

Now, consider the equation

$$\begin{aligned} [(\mathbf{I}_n - \frac{1}{n} \mathbf{M}) \otimes \mathbf{I}_p] \mathbf{V} [(\mathbf{I}_n - \frac{1}{n} \mathbf{M}) \otimes \mathbf{I}_p] &= (\mathbf{I}_n - \frac{1}{n} \mathbf{M}) \mathbf{W} (\mathbf{I}_n - \frac{1}{n} \mathbf{M}) \\ &\quad \otimes (n-1)^{-1} \boldsymbol{\Sigma}. \end{aligned} \quad (3.6)$$

If $\mathbf{W}$ is defined as in (3.5), then Part (1) of Lemma 4 follows when one notes that the general n.d. IDP covariance matrix $\mathbf{V}$ defined in (3.1) is a representation of the general n.d. solution to (3.6). The sufficiency portion of the proof concludes when we note that $\text{rank}[(\mathbf{I}_n - \frac{1}{n} \mathbf{M})] = n-1$.

To prove necessity, assume that $\mathbf{S}$ is defined as in (1.1) and that $\mathbf{S} \sim \mathbf{W}_p(n-1, (n-1)^{-1} \boldsymbol{\Sigma}, \triangle)$. Also, let $\boldsymbol{\Sigma} \in \mathbb{R}_p^{\geq}$ and note that

$(\mathbf{I}_n - \frac{1}{n}\mathbf{M})$ is idempotent. In the constraint equations (1), (2) and (3) in Lemma 4, let $\mathbf{A} = (n-1)^{-1} (\mathbf{I}_n - \frac{1}{n}\mathbf{M})$ and $\mathbf{W} = (n-1) (\mathbf{I}_n - \frac{1}{n}\mathbf{M})$. Then (1) in Lemma 4 is

$$\left[\left(\mathbf{I}_n - \frac{1}{n}\mathbf{M} \right) \otimes \mathbf{I}_p \right] \mathbf{V} \left[\left(\mathbf{I}_n - \frac{1}{n}\mathbf{M} \right) \otimes \mathbf{I}_p \right] = \left(\mathbf{I}_n - \frac{1}{n}\mathbf{M} \right) \otimes \Sigma, \quad (3.7)$$

and equations (2) and (3) in Lemma 4 constraining $\mathbf{W}$ are satisfied. Applying Lemma 3 to determine the general n.d. solution to equation (3.7) for $\mathbf{V}$, we have that

$$\begin{aligned} \mathbf{V} &= \left\{ \left(\mathbf{I}_n - \frac{1}{n}\mathbf{M} \right) \otimes \Sigma^{\frac{1}{2}} + \left[\mathbf{I}_{np} - \left(\mathbf{I}_n - \frac{1}{n}\mathbf{M} \right) \otimes \mathbf{I}_p \right] \mathbf{Z} \right\} \\ &\quad \cdot \left\{ \left(\mathbf{I}_n - \frac{1}{n}\mathbf{M} \right) \otimes \Sigma^{\frac{1}{2}} + \left[\mathbf{I}_{np} - \left(\mathbf{I}_n - \frac{1}{n}\mathbf{M} \right) \otimes \mathbf{I}_p \right] \mathbf{Z} \right\}' \\ &= \left(\mathbf{I}_n - \frac{1}{n}\mathbf{M} \right) \otimes \Sigma + \left[\left(\mathbf{I}_n - \frac{1}{n}\mathbf{M} \right) \otimes \Sigma^{\frac{1}{2}} \right] \left[\mathbf{Z}' \left(\frac{1}{n}\mathbf{M} \otimes \mathbf{I}_p \right) \right] \\ &\quad + \left[\left(\frac{1}{n}\mathbf{M} \otimes \mathbf{I}_p \right) \mathbf{Z} \right] \left[\left(\mathbf{I}_n - \frac{1}{n}\mathbf{M} \right) \otimes \Sigma^{\frac{1}{2}} \right] + \left(\frac{1}{n}\mathbf{M} \otimes \mathbf{I}_p \right) \mathbf{Z} \mathbf{Z}' \\ &\quad \cdot \left(\frac{1}{n}\mathbf{M} \otimes \mathbf{I}_p \right) \\ &= \left(\mathbf{I}_n - \frac{1}{n}\mathbf{M} \right) \otimes \Sigma + (\mathbf{M} \otimes \mathbf{I}_p) \mathbf{H} + \mathbf{H}' (\mathbf{M} \otimes \mathbf{I}_p), \end{aligned}$$

where $\mathbf{H} = \frac{1}{n}\mathbf{Z} \left[\left(\mathbf{I}_n - \frac{1}{n}\mathbf{M} \right) \otimes \Sigma^{\frac{1}{2}} + \frac{1}{2}\mathbf{Z}' \left(\frac{1}{n}\mathbf{M} \otimes \mathbf{I}_p \right) \right]$ and $\mathbf{Z} \in \mathbb{R}_{np \times np}$ is arbitrary.

The first of the two main n.d. IDP dependency structure characterizations for the sample covariance matrix is given below as a corollary. The corollary characterizes the general n.d. IDP covariance structure and mean matrix of a matrix-normal random matrix such that $\mathbf{S} \sim W_p \left(n-1, (n-1)^{-1} \Sigma \right)$,

where $\Sigma \in \mathbb{R}_p^{\geq}$. Our new result generalizes the sufficient p.d. IDP covariance structure for the sample covariance matrix given by Morrison (1962) and also generalizes the p.d. IDP characterization results of Pavur (1987) and Young, Meaux and Seaman (1994) to the n.d. case.

Corollary 1. Let $\mathbf{Y}_{n \times p} \sim \text{MN}(\boldsymbol{\mu}, \mathbf{V})$, where $\boldsymbol{\mu} \in \mathbb{R}_{n \times p}$ and $\mathbf{V} \in \mathbb{R}_{np}^{\geq}$, and let $\mathbf{S}$ be defined as in (1.1). Then, $\mathbf{S} \sim \mathbf{W}_p(n-1, (n-1)^{-1}\Sigma)$, where $\Sigma \in \mathbb{R}_p^{\geq}$, if and only if $\boldsymbol{\mu} = \mathbf{e}_n \otimes \boldsymbol{\nu}'$, where $\boldsymbol{\nu} \in \mathbb{R}_{p \times 1}$ is arbitrary, and

$$\mathbf{V} = \left(\mathbf{I}_n - \frac{1}{n}\mathbf{M} \right) \otimes \Sigma + (\mathbf{M} \otimes \mathbf{I}_p) \mathbf{H} + \mathbf{H}' (\mathbf{M} \otimes \mathbf{I}_p), \quad (3.8)$$

where $\mathbf{H}$ is defined as in Theorem 1.

Proof. The proof is a direct result of Theorem 1 and the fact that a representation of the general solution to $(\mathbf{I}_n - \frac{1}{n}\mathbf{M})\boldsymbol{\mu} = \mathbf{0}$ is $\boldsymbol{\mu} = \mathbf{e}_n \otimes \boldsymbol{\nu}'$, where $\boldsymbol{\nu} \in \mathbb{R}_{p \times 1}$ is arbitrary.

In the following corollary we characterize the general n.d. observation covariance matrix and mean vector of a normal random vector such that a multiple of the univariate sample variance is distributed as a central chi-squared random variable. The new n.d. IDP covariance matrix characterization extends the work of Baldessari (1965), Baldessari and Gallo (1981), Stadje (1984) and Jensen (1989) from the p.d. to the more general n.d. case. The corollary also simplifies a n.d. IDP characterization result for the univariate sample variance given in Young, Lehman and Meaux (1996).

Corollary 2. Let $\mathbf{Y}_{n \times 1} \sim \text{N}(\boldsymbol{\delta}, \mathbf{V})$, where $\boldsymbol{\delta} \in \mathbb{R}_{n \times 1}$ and $\mathbf{V} \in \mathbb{R}_n^{\geq}$, and let $\mathbf{S}^2 \equiv \frac{\mathbf{Y}'(\mathbf{I}_n - \frac{1}{n}\mathbf{M})\mathbf{Y}}{n-1}$. Then, $\frac{(n-1)}{\sigma^2}\mathbf{S}^2 \sim \chi_{n-1}^2$ if and only if $\boldsymbol{\delta} = m\mathbf{e}_n$, where m is an arbitrary scalar, and $\mathbf{V} = \sigma^2 (\mathbf{I}_n - \frac{1}{n}\mathbf{M}) + \mathbf{M}\mathbf{H}_u + \mathbf{H}_u'\mathbf{M}$,

where $\mathbf{H}_u = \frac{1}{n} \mathbf{Z} \left[\sigma \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) + \frac{1}{2n} \mathbf{Z}' \mathbf{M} \right]$, $\mathbf{Z} \in \mathbb{R}_{n \times n}$ is arbitrary, and $\sigma > 0$.

Proof. The proof follows from Corollary 1.

We now derive the second of our two main results. Assuming $\mathbf{Y}_{n \times p} \sim \text{MN}(\boldsymbol{\mu}, \mathbf{V})$, we derive a characterization of the general n.d. IDP covariance structure $\mathbf{V} \in \mathbb{R}_{np}^{\geq}$ such that the sample covariance matrix $\mathbf{S} \sim \mathbf{W}_p \left(n-1, (n-1)^{-1} \boldsymbol{\Sigma}, \Delta \right)$ and $\mathbf{S}$ is independent of $\bar{\mathbf{Y}}$, the sample mean vector. That is, we characterize the general n.d. IDP covariance structure of a matrix-normal random matrix such that the distribution of $\mathbf{S}$, defined in (1.1), is distributed exactly as if the observations have the i.i.d. covariance structure $\mathbf{v} = \mathbf{I}_n \otimes \boldsymbol{\Sigma}$ and $\mathbf{S}$ retains its independence from $\bar{\mathbf{Y}}$. We first prove the following theorem.

Theorem 2. Let $\mathbf{Y}_{n \times p} \sim \text{MN}(\boldsymbol{\mu}, \mathbf{V})$, where $\boldsymbol{\mu} \in \mathbb{R}_{n \times p}$ and $\mathbf{V} \in \mathbb{R}_{np}^{\geq}$, let $\mathbf{S}$ be defined as in (1.1), and let $\bar{\mathbf{Y}} \in \mathbb{R}_{p \times 1}$ be the sample mean vector. Then, $\mathbf{S}$ is independent of $\bar{\mathbf{Y}}$ and $\mathbf{S} \sim \mathbf{W}_p \left(n-1, (n-1)^{-1} \boldsymbol{\Sigma}, \Delta \right)$, where $\boldsymbol{\Sigma} \in \mathbb{R}_p^{\geq}$ and $\Delta = (n-1)^{-1} \boldsymbol{\mu}' \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \boldsymbol{\mu}$ if and only if

$$\mathbf{V} = \mathbf{I}_n \otimes \boldsymbol{\Sigma} + \mathbf{M} \otimes \mathbf{E}, \quad (3.9)$$

where $\mathbf{E} = \mathbf{Q} - \frac{1}{n} \boldsymbol{\Sigma}$ and $\mathbf{Q}, \boldsymbol{\Sigma} \in \mathbb{R}_p^{\geq}$ are arbitrary.

Proof. Assume $\mathbf{V}$ is defined as in (3.9), let $\mathbf{W}$ be defined as in (3.5), and note that $\bar{\mathbf{Y}} \bar{\mathbf{Y}}' = \frac{1}{n} \mathbf{Y}' \mathbf{M} \mathbf{Y}$. Sufficiency follows from the fact that the n.d. matrix $\mathbf{V}$ simultaneously satisfies the pair of matrix equations

$$\left[\left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right] \mathbf{V} \left[\left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right] = \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \mathbf{W} \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes (n-1)^{-1} \boldsymbol{\Sigma}$$

and

$$\left[(n-1)^{-1} \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right] \mathbf{V} \left(\frac{1}{n} \mathbf{M} \otimes \mathbf{I}_p \right) = \mathbf{0},$$

where $\mathbf{W}$ is given in (3.5), from Lemma 4 and the fact that functions of independent random variables are independent.

To prove necessity, assume that $\mathbf{S}$ is defined as in (1.1) such that $\mathbf{S} \sim \mathbf{W}_p \left(n-1, (n-1)^{-1} \boldsymbol{\Sigma}, \triangle \right)$ and that $\mathbf{S}$ is independent of $\bar{\mathbf{Y}}$. Note that $\bar{\mathbf{Y}} = \frac{1}{n} \mathbf{Y}' \mathbf{e}_n$ and $\frac{1}{n^2} \mathbf{e}_n \mathbf{e}_n' = \frac{1}{n} \mathbf{M}$. Because $\mathbf{S}$ and $\bar{\mathbf{Y}}$ are independent, we have from (4) of Lemma 4 that

$$\left[\left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right] \mathbf{V} \left(\frac{1}{n} \mathbf{M} \otimes \mathbf{I}_p \right) = \mathbf{0}. \quad (3.10)$$

From (3.10) we therefore have that

$$\begin{aligned} \left[\left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right] \mathbf{V} \left(\frac{1}{n} \mathbf{M} \otimes \mathbf{I}_p \right) &= \left[\left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right] \\ &\quad \mathbf{V} \left[\left(\mathbf{I}_n - \mathbf{I}_n + \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right] \\ &= \left[\left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right] \mathbf{V} - \\ &\quad \left[\left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right] \\ &\quad \cdot \mathbf{V} \left[\left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right] \\ &= \mathbf{0}. \end{aligned}$$

Hence, $\left[\left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right] \mathbf{V} = \left[\left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right] \mathbf{V} \left[\left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right]$. Utilizing constraint equation (1) of Lemma 4 with $\mathbf{W}$ defined as in (3.5), we have that $\mathbf{V}$ must satisfy

$$\left[\left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right] \mathbf{V} = \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \boldsymbol{\Sigma}. \quad (3.11)$$

The application of Lemma 5 to equation (3.11) and some algebraic simplification provides a representation of the general n.d. solution to (3.11) of the form

$$\mathbf{V} = \mathbf{I}_n \otimes \boldsymbol{\Sigma} + \mathbf{M} \otimes \mathbf{E},$$

where $\mathbf{E} = \mathbf{Q} - \frac{1}{n}\boldsymbol{\Sigma}$, and $\mathbf{Q} \in \mathbb{R}_p^{\geq}$ is arbitrary.

We now characterize the n.d. matrix-normal observation covariance matrix and mean matrix such that $\mathbf{S} \sim \mathbf{W}_p(n-1, (n-1)^{-1}\boldsymbol{\Sigma})$ and $\mathbf{S}$ is independent of $\bar{\mathbf{Y}}$. The following corollary is a multivariate extension of p.d. IDP results for the univariate sample variance and sample mean by Bhat (1962), Rogers (1980) and Stadje (1984). The corollary also is a generalization of a p.d. IDP general covariance structure result for the multivariate sample covariance matrix and sample mean vector by Pavur (1987) and Young, Meaux and Seaman (1994). In addition, the corollary is an extension of an n.d. IDP covariance matrix result derived in Khatri (1963) under the restriction $\mathbf{V} = \mathbf{U} \otimes \boldsymbol{\Sigma}$, where $\mathbf{U} \in \mathbb{R}_n^{\geq}$ and $\boldsymbol{\Sigma} \in \mathbb{R}_p^{\geq}$.

Corollary 3. Let $\mathbf{Y}_{n \times p} \sim \text{MN}(\boldsymbol{\mu}, \mathbf{V})$, where $\boldsymbol{\mu} \in \mathbb{R}_{n \times p}$ and $\mathbf{V} \in \mathbb{R}_{np}^{\geq}$, and let $\mathbf{S}$ be defined as in (1.1). Then, $\mathbf{S} \sim \mathbf{W}_p(n-1, (n-1)^{-1}\boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma} \in \mathbb{R}_p^{\geq}$, and $\mathbf{S}$ is independent of $\bar{\mathbf{Y}}$ if and only if $\boldsymbol{\mu} = \mathbf{e}_n \otimes \boldsymbol{\nu}'$, where $\boldsymbol{\nu} \in \mathbb{R}_{p \times 1}$ is arbitrary, and

$$\mathbf{V} = \mathbf{I}_n \otimes \boldsymbol{\Sigma} + \mathbf{M} \otimes \mathbf{E}, \quad (3.12)$$

where $\mathbf{E} = \mathbf{Q} - \frac{1}{n}\boldsymbol{\Sigma}$ and $\mathbf{Q} \in \mathbb{R}_p^{\geq}$ is arbitrary.

Proof. The proof follows immediately from Theorem 2 and the fact that a representation of the general solution to $(\mathbf{I}_n - \frac{1}{n}\mathbf{M})\boldsymbol{\mu} = \mathbf{0}$ is $\mathbf{e}_n \otimes \boldsymbol{\nu}'$, where $\boldsymbol{\nu} \in \mathbb{R}_{p \times 1}$ is arbitrary.

In the last corollary we characterize the covariance matrix and mean vector of a normal random vector such that a multiple of a univariate sample variance is distributed as a central chi-squared random variable and is independent of the sample mean. The new general n.d. IDP univariate-observation covariance matrix characterization generalizes the p.d. IDP characterization result for the univariate sample variance given in Bhat (1962), Rogers (1980) and Stadje (1984).

Corollary 4. Let $\mathbf{Y}_{n \times 1} \sim N(\delta, \mathbf{V})$, where $\delta \in \mathbb{R}_{n \times 1}$ and $\mathbf{V} \in \mathbb{R}_p^{\geq}$, and let $S^2 \equiv \frac{\mathbf{Y}'(\mathbf{I} - \frac{1}{n}\mathbf{M})\mathbf{Y}}{n-1}$. Then, $\frac{(n-1)}{\sigma^2}S^2 \sim \chi_{n-1}^2$ and S^2 is independent of $\bar{\mathbf{Y}}$ if and only if $\delta = m\mathbf{e}_n$, where m is a scalar, and $\mathbf{V} = \sigma^2\mathbf{I}_n + \lambda\mathbf{M}$, where $\lambda = \gamma - \frac{1}{n}\sigma^2$ with $\gamma \geq 0$.

Proof. The proof follows from Corollary 3.

4. Comments

We have explicitly characterized the general n.d. IDP covariance of a matrix normal random matrix for the multivariate sample covariance matrix $\mathbf{S}$ defined in (1.1). In doing so, we have also given a simpler representation of the general n.d. IDP covariance structure for a multiple of the univariate sample variance. We remark that the mean matrix given in Corollary 1 is the mean matrix of the observations under the i.i.d. assumption. We also note that the general n.d. IDP covariance matrix characterization result for $\mathbf{S}$ given in Theorem 1 can also be obtained as a special case of the general n.d. IDP covariance structure characterizations for families of multivariate quadratic forms formulated by Lehman (1993) and Wong and Masaro (1999). However, our general n.d. IDP covariance structure characterization result for $\mathbf{S}$ is derived in a more direct and concise manner.

One question of interest concerns the relationship between our characterization of the general n.d. IDP covariance structure given in (3.8) and the characterization of the general p.d. IDP covariance structure given in Young, Meaux and Seaman (1994). We can easily show that the general n.d. IDP dependency structure given in (3.8) contains the general p.d. IDP covariance matrix derived in Young, Meaux and Seaman (1994) in that one need only let $\mathbf{Z}$ be a matrix simultaneously satisfying the two conditions $\mathcal{N}(\mathbf{Z}') \cap \mathcal{N}[(\mathbf{I}_n - \frac{1}{n}\mathbf{M}) \otimes \mathbf{I}_p] = \{0\}$ and $\mathcal{R}[(\mathbf{I}_n - \frac{1}{n}\mathbf{M}) \otimes \Xi^{\frac{1}{2}}] \cap \mathcal{R}[\mathbf{Z}'(\mathbf{M} \otimes \mathbf{I}_p)] = \{0\}$.

We have also characterized the general n.d. IDP covariance structure of a matrix-normal random matrix such that the sample covariance matrix is distributed as a noncentral Wishart random matrix and is independent of the sample mean. As a consequence, we have also given a representation of the general n.d. IDP covariance structure such that a multiple of the univariate sample variance is distributed as a chi-squared random variable and is independent of the univariate sample mean. We again note that the mean matrix given in Corollary 3 is the i.i.d. mean matrix.

Another question of interest concerns the relationship between our characterization of the general n.d. IDP covariance structure given in (3.12) and the characterization of the general p.d. IDP covariance structure derived in Young, Meaux and Seaman (1994). If in (3.12) one lets $\mathbf{E} = \mathbf{D} - \frac{1}{n}\mathbf{\Sigma}$, where $\mathbf{\Sigma}, \mathbf{D} \in \mathbb{R}_p^{\geq}$ is arbitrary, then we have that

$$\begin{aligned} \mathbf{V} &= \mathbf{I}_n \otimes \mathbf{\Sigma} + \mathbf{M} \otimes \mathbf{E}, \\ &= \mathbf{I}_n \otimes \mathbf{\Sigma} + \mathbf{M} \otimes \left(\mathbf{D} - \frac{1}{n}\mathbf{\Sigma} \right) \\ &= \mathbf{I}_n \otimes \mathbf{\Sigma} - \frac{1}{n}\mathbf{M} \otimes \mathbf{\Sigma} + \mathbf{M} \otimes \mathbf{D} \end{aligned}$$

$$= \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \boldsymbol{\Sigma} + \mathbf{M} \otimes \mathbf{D}.$$

Hence, the general n.d. IDP dependency structure given in (3.12) contains the general p.d. IDP covariance structure derived in Young, Meaux and Seaman (1994). We also comment that the general IDP covariance structure (3.12) is contained in the general IDP covariance structure characterization derived by Young, Odell and Seaman (1994) for the independence of two multivariate quadratic forms.

Finally, we emphasize that the parameter matrix $\boldsymbol{\Sigma}$ in the representations (3.8) and (3.12) need not be p.d., such as in previous IDP covariance structure characterizations corresponding with $\mathbf{S}$. Therefore, the two IDP observation covariance matrices explicitly characterized in (3.8) and (3.12) are the most general IDP covariance structures possible for the two cases considered in this paper.

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